

Applied NLP

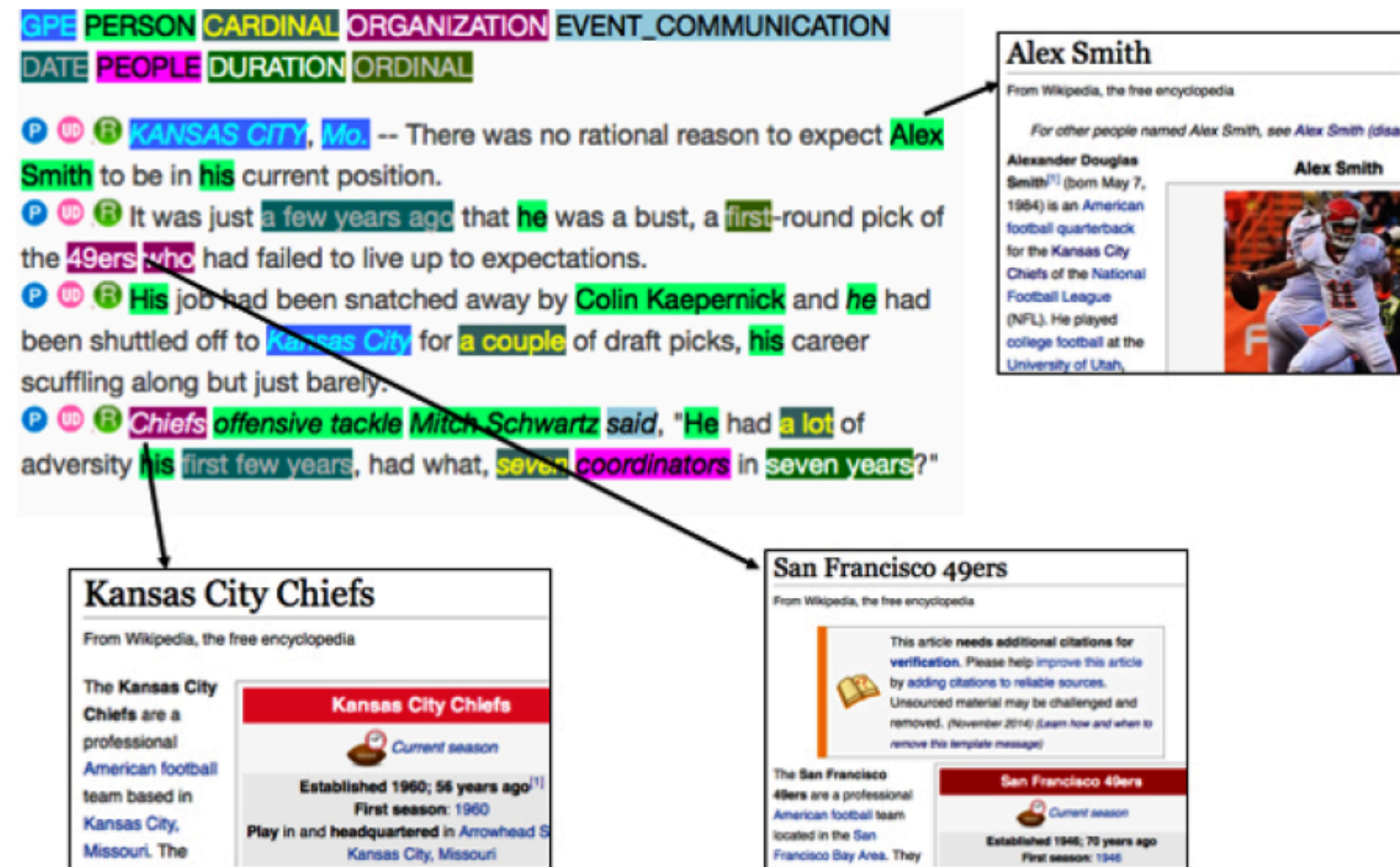
DATA 641

Lecture 11 — Other NER Applications

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

Named Entity Disambiguation and Linking

- Consider a scenario where we are working on the data science team of a large newspaper publication (say, The New York Times). We are charged with the task of building a system that creates visual representation of news stories by connecting different entities mentioned in the stories to what they refer to in the real world.

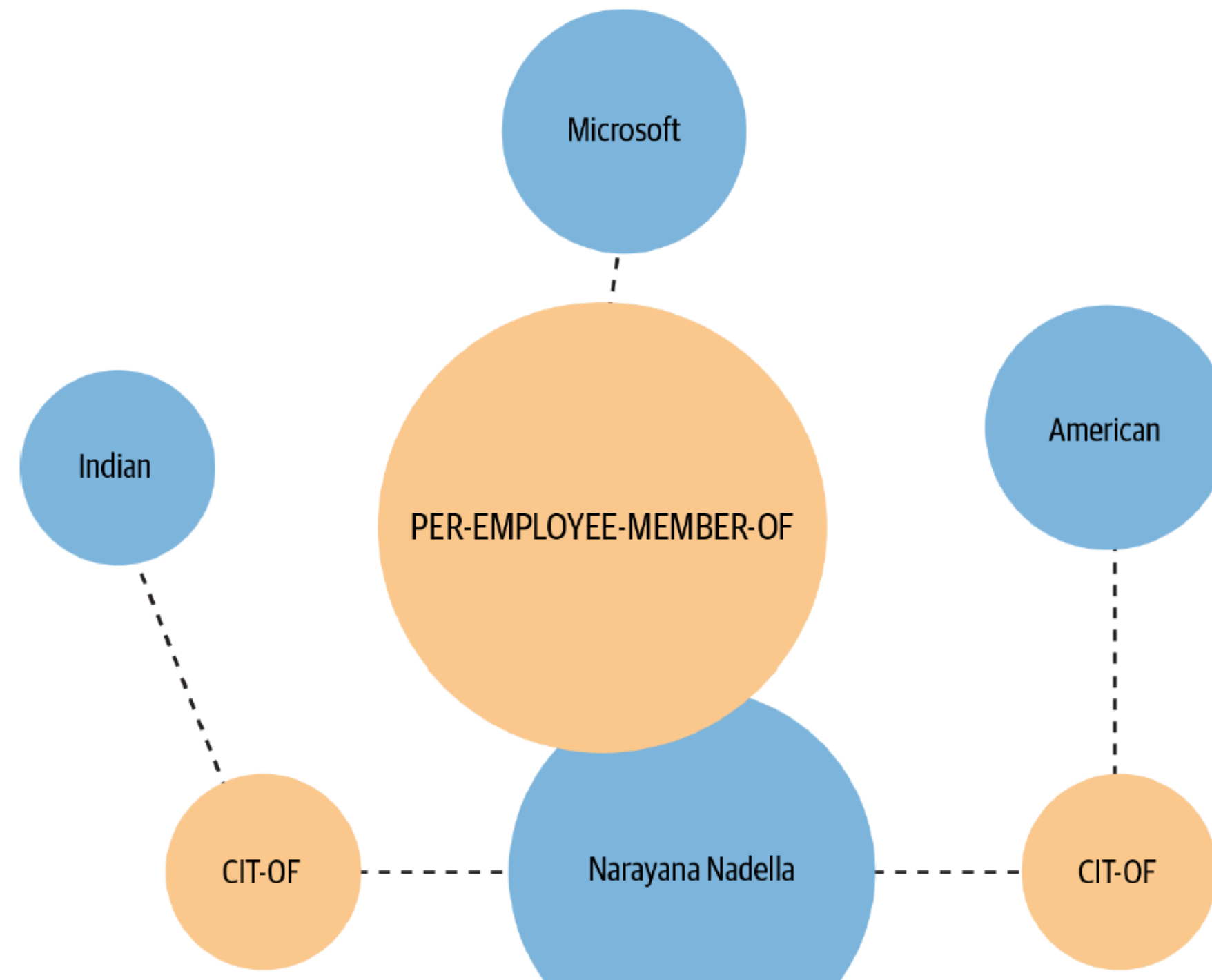


- Doing this requires knowledge of several IE tasks beyond what we have seen with NER and KPE.

- Named entity disambiguation (NED) refers to the NLP task of assigning a unique identity to entities mentioned in the text. NER and NED together are known as named entity linking (NEL). Some other NLP applications that would need NEL include question answering and constructing large knowledge bases of connected events and entities.
- To build an Information Extraction (IE) system for Named Entity Linking (NEL), context is crucial, similar to Named Entity Recognition (NER). However, NEL requires more extensive NLP pre-processing. It involves parsing to identify linguistic elements like subjects, verbs, and objects, and may also need coreference resolution to link multiple references to the same entity.
- State-of-the-art NEL utilizes various neural network architectures and relies on large annotated datasets and encyclopedic resources for linking. NEL is a specialized NLP task, and in practice, many prefer using off-the-shelf services from providers like IBM (Watson) and Microsoft (Azure) instead of developing in-house systems.

Relationship Extraction

- Relationship extraction (RE) is the IE task that deals with extracting entities and relationships between them from text documents. It is an important step in building a knowledge base, and it is also useful in improving search and developing question-answering systems.



Approaches to RE

- In the field of NLP, RE is an extensively studied area. Researchers have investigated various methods, ranging from manual pattern creation to supervised, semi-supervised, and unsupervised learning approaches, in the development of RE systems.
- Hand-crafted patterns, including regular expressions, have been employed to capture specific relationships. While these hand-built patterns offer high precision, they often suffer from limited coverage and the challenge of encompassing all possible relations within a given domain.
- Hence, RE is often treated as a supervised classification problem. The datasets used to train RE systems contain a set of pre-defined relations, similar to classification datasets. This consists of modeling it as a two-step classification problem:
 - Whether two entities in a text are related (binary classification).
 - If they are related, what is the relation between them (multiclass classification)?
- When it's not possible to obtain training data for supervised approaches, unsupervised methods come into play. Unsupervised Relation Extraction (also referred to as "open IE") is designed to extract relationships directly from the web, without depending on any pre-defined training data or a predefined list of relations.